

EXERCISE 4: SQL DATE FUNCTIONS

PROBLEM ZINDONE

Q1: Return each order with the day name of the date

```
SELECT order_id,  
       customer_name,  
       order_date,  
       DAYNAME(order_date) AS day_name  
FROM orders;
```

Output

order_id	customer_id	order_date	day_name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

Q2: Return each customer with the name of the month in which they signed up.

```
SELECT customer_id,  
       customer_name,  
       signup_date,  
       MONTHNAME(signup_date) AS signup_month_name  
FROM customer_signup
```

Output

customer_id	customer_name	signup_date	signup_month_name
201	John	2026-01-15	January
202	Mary	2026-02-10	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

Q3. Return each sale with the month number extracted from sales_date.

```
SELECT sales_id,  
       product_name,  
       sale_date,  
       MONTH (sale_date) AS sale_month  
FROM sales;
```

Output

sale_id	product_name	sale_date	sale_month
SOD1	Laptop	2026-01-10	1
SOD2	Mouse	2026-02-15	2
SOD3	Keyboard	2026-03-20	3
SOD4	Monitor	2026-04-25	4
SOD5	Desk	2026-05-30	5

Q4. Return each transaction with the year extracted from transaction_date

```
SELECT transaction_id,  
       customer_id,  
       transaction_date,  
       YEAR (transaction_date) AS transaction_year  
FROM transactions;
```

Output

transaction_id	customer_id	transaction_date	transaction_year
TOD1	101	2024-12-15	2024
TOD2	102	2025-01-20	2025
TOD3	103	2025-06-10	2025
TOD4	104	2026-02-05	2026
TOD5	105	2026-05-25	2026

Q5. Return each delivery with the day of the month extracted from delivery_date

```
SELECT delivery_id,
       customer_id,
       delivery_date,
       DAY (delivery_date) AS day_of_month
FROM deliveries;
```

Output

delivery_id	customer_id	delivery_date	day_of_month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2025-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Q6. Return all employees and add a column showing today's date

```
SELECT employee_id,
       employee_name,
       department,
       CURRENT_DATE ( ) AS today_date
```

FROM employees;

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Output

employee_id	employee_name	employee name	department	today_date
301	Nandi	Nandi	Sales	2026-06-17
302	Brian	Brian	IT	2026-06-17
303	Lerato	Lerato	Finance	2026-06-17
304	Sipho	Sipho	HR	2026-06-17
305	Aisha	Aisha	Marketing	2026-06-17

Q7 Convert the order_date_text string column into a proper date column

```
SELECT order_id,
       customer_id,
       order_date_text,
       TO_DATE (order_date_text) AS order_date
FROM online_orders;
```

Output

order_id	customer_id	order_date_text	order_date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

Q8 Format payment_date as a text string in the format YYYY-MM-DD.

```
SELECT payment_id,
       customer_id,
       payment_date,
       TO_CHAR (payment_date, 'YYYY-MM-DD') AS formatted_payment_date
FROM order payment_dates;
```

Output

payment_id	customer_id	payment_date	formatted_payment_date
POD1	101	2026-01-05	2026-01-05
POD2	102	2026-02-10	2026-02-10
POD3	103	2026-03-15	2026-03-15
POD4	104	2026-04-20	2026-04-20
POD5	105	2026-05-05	2026-05-25

Q9. Calculate how many days have passed since each customer's last purchase. Use today's date

```
SELECT customer_id,  
       customer_name,  
       last_purchase_date,  
       DATEDIFF(CURRENT_DATE(), last_purchase_date)  
       AS days_since_last_purchase  
FROM customer_purchases;
```

Output

customer_id	customer_name	last_purchase_date	days_since_last_purchase
501	John	2026-05-01	47
502	Mary	2026-05-10	38
503	Peter	2026-05-15	33
504	Sarah	2026-05-20	28
505	Thabo	2026-05-25	23

Q10. Calculate the expected delivery date by adding 7 days to the order_date.

```
SELECT order_date,  
       customer_id,  
       order_date,  
       DATEADD(order_date, 7) AS expected_delivery_date  
FROM shipping_orders;
```

Output

order_id	customer_id	order_date	expected_delivery_date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Q11 Extract the year, month number and day from booking date into three separate columns

```
SELECT booking-id,  
        customer-id,  
        booking-date,  
        YEAR(booking-date) AS booking-year,  
        MONTH(booking-date) AS booking-month,  
        DAY(booking-date) AS booking-day  
FROM bookings;
```

Output

booking-id	customer-id	booking-date	book-year	book-month	book-day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

Q12

```
SELECT order-id,  
        customer-id,  
        order-date,  
        YEAR(order-date) = 2026 AS order-year,  
        amount
```

```
FROM yearly-orders  
WHERE YEAR(order-date) = 2026;
```

Outcome

order-id	customer-id	order-date	order-year	amount
7004	104	2026-02-05	True	1500
7005	105	2026-05-25	True	2000

Q13 Return only orders placed in March (month number 3)

```
SELECT order_id,  
       customer_id,  
       order_date,  
       MONTH(order_date) = 3 AS order_month,  
       amount
```

```
FROM monthly_orders
```

```
WHERE MONTH(order_date) = 3;
```

Outcome

order_id	customer_id	order_date	order_month	amount
8003	103	2026-03-10	True	800
8004	104	2026-03-25	True	1500

Q14 Return the last day of the month for each subscription start date

```
SELECT subscription_id,  
       customer_id,  
       start_date,  
       LAST_DAY(start_date) AS month_end_date  
FROM subscriptions;
```

Outcome

subscription_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

Q15. Return the first day of the month for each campaign send date

```
SELECT send_id
       customer_id
       send_date
       DATE_TRUNC('MONTH', send_date) AS month_start_date
FROM campaign_sends;
```

Output

send_id	customer_id	send_date	month_start_date
COD1	101	2026-01-12	2026-01-01
COD2	102	2026-02-18	2026-02-01
COD3	103	2026-03-22	2026-03-01
COD4	104	2026-04-09	2026-04-01
COD5	105	2026-05-27	2026-05-01

Q16 Format invoice_date as a text string showing month name and year for e.g January 2026

```
SELECT invoice_id,
       customer_id,
       invoice_date,
       TO_CHAR(invoice_date, 'MMMM, YYYY') AS
       invoice_date-month invoice_month_date
FROM 'invoice_dates';
```

Output

invoice_id	customer_id	invoice_date	invoice_month_date
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-25	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

Q17 Calculate each customers age in years using their date of birth and today's date.

```
SELECT customer_id,  
       customer_name,  
       date-of-birth,  
       FLOOR(DATEIFF(CURRENT-DATE(), date-of-birth  
                     /365.25)) AS customer-age  
FROM customer birthdays;
```

Output

customer_id	customer_name	date-of-birth	customer_age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thabo	2000-07-30	25

Q18 Classify days based on day name of their order date.

```
SELECT order_id,  
       customer_id,  
       order-date,  
       DAYNAME(order-date) AS day-name,  
       CASE  
         WHEN DAYOFWEEK(order-date) IN (1, 7) THEN  
           'weekend'  
         ELSE 'weekday'  
       END AS day-type  
FROM weekend orders;
```


Q18 continued

output

order_id	customer_id	order_date	day_name	day_type
9001	101	2026-05-01	Friday	Weekday
9002	102	2026-05-02	Saturday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekday
9005	105	2026-05-05	Tuesday	Weekday

Q19 Extract the quarter number from each transaction_date

```

SELECT transaction_id,
       customer_id,
       transaction_date,
       DATE_FORMAT(transaction_date, 'Q') AS transaction_
       quarter,
       amount
FROM quarterly_transactions;

```

Output

transaction_id	customer_id	order_date	transaction_ quarter	amount
Q001	101	2026-01-15	Q1	500
Q002	102	2026-03-20	Q1	1200
Q003	103	2026-04-10	Q2	800
Q004	104	2026-07-05	Q3	1500
Q005	105	2026-10-25	Q4	2000

Q20 Return days more than 30 days from today

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DATEDIFF (CURRENT_DATE (), order_date) AS days-since-order  
       amount  
FROM recent orders  
WHERE DATEDIFF (CURRENT_DATE (), order_date) > 30;
```

Output

order_id	customer_id	order_date	Day-since-order	amount
R001	101	2026-04-01	78	500
R002	102	2026-04-15	64	1200
R003	103	2026-05-01	48	800
R004	104	2026-05-10	39	1500

Q21 Customer recency

```
SELECT customer_id,  
       customer_name,  
       last_purchase_date,  
       DATEDIFF(CURRENT_DATE(), last_purchase_date) AS  
       day-since-last-purchase,  
       CASE  
           WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date)  
               <= 30 THEN 'Active'  
           WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date)  
               BETWEEN 31 AND 90 THEN 'At Risk Customer'  
           WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date)  
               > 90 THEN 'Inactive Customer'  
       END AS customer_status  
FROM customer_recency;
```

Output

customer_id	customer_name	last_purchase_date	day-since-last-purchase	customer_status
101	John	2026-05-25	24	Active
102	Mary	2026-05-10	39	At Risk Customer
103	Peter	2026-04-01	78	At Risk Customer
104	Sarah	2026-02-15	123	Inactive Customer
105	Thabo	2025-12-20	180	Inactive Customer